Chapter 9 - Gibbs sampling: Exercise solutions

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Click on the arrow to see a solution.

Exercise 9.1

NOT COMPLETE!

The dataset bivarmissing.csv contains $n = 20$ observations on the two variables X_1 and X_2. Note that four observations are missing in the dataset (coded with NA). Assume the bivariate normal model for the data: $\mathbf{x}_1, ..., \mathbf{x}_n | \theta, \Sigma \overset{iid}{\sim} \mathrm{N}(\theta, \Sigma)$, with mean vector $\theta = (\theta_1, \theta_2)^\top$ and covariance matrix Σ.

a) We first assume that the covariance matrix is known to be

$$\Sigma = \begin{pmatrix} 1 & 1.5 \\ 1.5 & 4 \end{pmatrix}.$$

Let the complete dataset X be partitioned in observed and missing parts, $X = (X_{\text{obs}}, X_{\text{mis}})$. Implement a Gibbs sampling algorithm with two update steps:

1. simulate the mean vector θ conditional on all data, X.
2. simulate the missing data X_{mis} conditional on the observed data X_{obs} and the mean vector θ.

Simulate 7000 iterations of the Gibbs sampler, discard the first 2000 as burn-in and plot the marginal posterior distributions of θ_1 and θ_2. Use the prior $\theta \sim N(0, 10^2 I)$. Compare with the posterior obtained by simply discarding observations where at least one of the two variables have missing values.

Hint: use the formulas for the conditional distributions in a bivariate normal in Step 2 above.

Solution

Let us start by loading the data and locate the missing values

```
data = read.csv(file = "https://github.com/mattiasvillani/BayesianLearningBook/raw/main/dat
                header = TRUE, row.names = NULL)
n = dim(data)[1]
# Finding the observations with at least on missing value
missing_obs = c()
for (i in 1:n){
  if (any(is.na(data[i,]))) missing_obs = c(missing_obs, i)
}
```

The update step for the mean vector conditional on all the data, including draws of the missing values, is given in Chapter 3 of the Bayesian Learning book:

$$\theta|x_{\text{obs}}, x_{\text{mis}} \sim N(\theta_n, \Lambda_n)$$

where

$$\begin{aligned}\theta_n &= (\Lambda_0^{-1} + n\Sigma^{-1})^{-1}(\Lambda_0^{-1}\theta_0 + n\Sigma^{-1}\bar{\mathbf{x}}) \\ \Lambda_n^{-1} &= {\Lambda_0}^{-1} + n\Sigma^{-1}.\end{aligned}$$

where prior mean $\theta_0 = 0$ and the prior covariance matrix $\Lambda_0 = 10^2 I$.
We implement this updating step as a separate function

```
Sigma = matrix(c(1, 1.5, 1.5, 4), 2, 2)
Lambda0 = 10^2*diag(2)
invSigma = solve(Sigma)
invLambda0 = solve(Lambda0)
library(mvtnorm) # We need rmvnorm to simulate from bivariate normal distr

UpdateMeanVector <- function(X, invSigma, theta0, invLambda0){
  n = dim(X)[1]
  xBar = colMeans(X) # sample mean of all data, including draws of missing
  Lambda_n  = solve(invLambda0 + n*invSigma) # post covariance
  theta_n = Lambda_n %*% n*invSigma %*% xBar # prior mean is zero
  thetaDraw = rmvnorm(1, theta_n, Lambda_n)
  return(as.vector(thetaDraws))
}
```

Next, we define a function that samples all the missing values

```
UpdateMissing <- function(X, missing_obs, data, theta, Sigma){
  for (i in missing_obs){
    misvar = which(is.na(data[i, ]))
    if (length(misvar) == 1){
      thetaCond = theta[misvar] +
        (Sigma[misvar,-misvar]/Sigma[-misvar,-misvar])*(data[,-misvar] - theta[-misvar])
      SigmaCond = Sigma[misvar,misvar] - Sigma[misvar,-misvar]^2/Sigma[-misvar, -misvar]
      X[i, misvar] = rnorm(1, thetaCond, SigmaCond)
    }else
    {
      X[i, ] = rmvnorm(1, theta, Sigma)
    }
  }
}
```

And finally we set define the complete Gibbs sampler

```
GibbsMultiNormalMissing <- function(data, theta0, Lambda0, Sigma, nSim){
  X = data
  n = dim(data)[1]
  # Finding the observations with at least on missing value
  missing_obs = c()
  for (i in 1:n){
    if (any(is.na(data[i,]))) missing_obs = c(missing_obs, i)
  }
  for (i in 1:nSim){

    # Update mean vector

    # Update missing values
  }
}
```

b) Assume now that Σ is unknown. We use the conjugate prior presented in Chapter 3 of the book

$$\theta|\Sigma \sim \mathrm{N}(\theta_0, \Sigma/\kappa_0) \tag{1}$$
$$\Sigma \sim \mathrm{IW}(\nu_0, \nu_0\Sigma_0), \tag{2}$$

where ν_0 is the degrees of freedom and Σ_0 is our best guess of Σ a priori. Use the non-informative choice $\nu_0 = 2$ and set

$$\Sigma_0 = \begin{pmatrix} 1 & 1.5 \\ 1.5 & 4 \end{pmatrix}.$$

Solution